Can one hidden-sector reading cover several dark-matter puzzles? Consistency checks of Ω_c/Ω_b = 3√3 against Planck and DESI DR2, the evolution of the RAR scale, and limits on uniform decay

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Abstract

This record asks whether one reading of dark matter suggested by the author's framework (Mathematical Materiality, MM) can connect several dark-matter puzzles, and confronts the parts that give numbers with public data. The cosmological ratio fit, the evolution of the radial-acceleration-relation (RAR) scale, the decay analysis and the hidden-sector constructions are separate investigations; no joint likelihood or complete dynamical model is presented. Hypothesis (declared, not derived): the effective gravitating cold-dark-matter density represents hidden-sector matter with its own internal interactions — in MM terms the ordinary matter of a «mirror», normal in its own laws, with constants read relative to ours, left generic — with the mean ratio ρ_c/ρ_b = 3√3 imposed separately. It is additionally postulated that hidden radiation does not source our expansion; no covariant implementation is provided.

Ugo Lanciano · Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635 · October 2026. Text and data: CC BY 4.0. Code: MIT licence. In the working records this deposit is called «materia oscura» (folder 29); some files carry the provisional name «MM 2.46» or the numbers of the working folders. They belong to this record.

In brief

Mathematical Materiality (MM) is the author's framework. This record asks whether one reading of dark matter suggested by MM can address several puzzles at once, and confronts the parts that give numbers with public data.

The cosmological ratio fit, the RAR evolution fit, the decay analysis and the hidden-sector constructions are separate investigations. **No joint likelihood or complete dynamical model connecting all of them is presented.**

The hypothesis, stated (section 6). We hypothesise that the effective gravitating cold-dark-matter density represents a contribution from hidden-sector matter with its own internal interactions, distinct from the visible baryons counted in ω_b. In MM terms this is the ordinary matter of a «mirror», normal in its own laws, whose constants are read relative to ours; its particle masses, dimensionless couplings and thermal history remain unspecified. We separately impose the cosmological mean-density ratio ρ_c/ρ_b = 3√3.

- The interpretation additionally postulates that hidden radiation does not source our cosmological expansion. This does not follow from electromagnetic separation; no covariant implementation is provided.
- Under standard universal gravitational coupling, a thermally identical copy of our sector is excluded by the radiation density (ΔN_eff ≈ 7.4); collisions, cooling and discs need separate modelling.
- A worked one-factor example satisfies two approximate screening inequalities (scattering and residual ionisation). It does not establish a viable hidden sector; a complete calculation is missing.

Consistency checks against data. `python riproduci.py` reports 745 successful checks. They cover algebraic identities, numerical calculations and consistency checks on archived cosmological outputs. The cosmological likelihood minimisations are not repeated by the script.

1. **The ratio Ω_c/Ω_b = 3√3 = 5.196.**
2. In flat ΛCDM with Σm_ν = 0.06 eV, imposing ω_cdm = 3√3·ω_b costs T_B = 0.17 in Q with Planck 2018 (lite TTTEEE, low-ℓ TT and EE) + DESI DR2 BAO, and 0.23 adding the Planck lensing reconstruction (CLASS + Cobaya).
3. The free fits give ω_c/ω_b = 5.22 and 5.21.
4. This is a retrospective consistency assessment using previously available data, not an independent prospective

test. A prospective test on specified future data is registered.
5. **The RAR acceleration scale increases with redshift in the four MUSE-DARK III bins.**
6. Among a constant, the Hubble rate and the inverse cosmic age, H(z) gives the lowest χ^2.
7. Its absolute fit quality depends strongly on how the plotted uncertainties are read: $\chi^2 = 5.48$ (bars as 1σ) or 21.04 (bars as 95 %), for three degrees of freedom.
8. Other simple forms do as well. Galaxies at $z \approx 2$ are in tension with it.
9. H(z) is kept as a phenomenological hypothesis and registered for future data.
10. **No preference for constant-rate decay of (almost) all dark matter into dark radiation** is found in the sampled profiles; their lowest value is at $\Gamma = 0$. At the conventional threshold $\Delta Q = 2.71$ the nominal, provisional bounds on the comoving loss are about 2.6 % (with the lensing reconstruction) and 3.8 % (without).
11. **Clusters.** The fixed ratio gives a baryon fraction of baryons plus cold dark matter of 0.161, a rough match to the hydrostatic baryon budgets of massive clusters. The galaxy RAR does not describe clusters (X-COP); that problem remains.

Problems this reading tries to connect (section 7): the dark-to-baryon ratio, the cosmological scale of a_0, the absence of direct-detection signals, and the cold-in-cosmology / RAR-in-galaxies behaviour. The connections have different statuses: imposed numerical hypotheses, phenomenological fits and qualitative assumptions. Other research programmes also address combinations of these puzzles. The case for the reading rests on this consilience, and it grows only as more links become quantitative.

How this work can turn out wrong (section 10): two quantitative hypotheses with predeclared failure thresholds for prospective tests.

Open:

- the dynamics of the hidden sector (thermal history, plasma, collisions);
- the Bullet cluster in detail;
- a_0 from the theory (the literal value $cH_0/2$ is 2.73 times the measured one);
- a derivation of the RAR;
- a covariant model in which hidden radiation does not gravitate.

1. What a physicist will recognise

- **ΛCDM fits with a fixed ratio ω_c/ω_b:** standard likelihoods and codes (Planck 2018, DESI DR2, CLASS, Cobaya); one linear constraint between two parameters.
- **The radial acceleration relation** (McGaugh, Lelli and Schombert 2016), g_obs = g_bar/(1 − exp(−√(g_bar/a_0))), and the question of its evolution (MUSE-DARK III; Alcock 2026). The closeness of a_0 to $cH_0/2\pi$ is an old remark by Milgrom.
- **Decaying dark matter into dark radiation:** CLASS «dcdm»; Audren et al. 2014, Poulin et al. 2016, Alvi et al. 2022, Bucko, Giri and Schneider 2023.
- **Hidden or mirror sectors whose ordinary matter appears as dark matter:**
- Kolb, Seckel and Turner 1985;
- mirror matter (Berezhiani, Foot, Ciarcelluti);
- the manyfold universe (Arkani-Hamed, Dimopoulos, Dvali and Kaloper);
- mass rescaling in Randall-Sundrum geometries;
- atomic dark matter (Kaplan et al. 2010; Cyr-Racine and Sigurdson 2013).

2. The ratio 3√3 (`codice/08_cosmology_results.py`, `registrazioni/rapporto_3r3/`)

Setup.

- **Model:** flat ΛCDM, one massive neutrino of 0.06 eV and N_ur = 2.0328 (N_eff ≈ 3.046, kept for comparison with CAMB), halofit.
- **Parameters:** both fits vary ln(10^{10}A_s), n_s, 100θ_s, ω_b, τ and A_planck. The free model also varies ω_cdm. D-B instead imposes ω_cdm = 3√3·ω_b.
- **Data:**
- PD = Planck 2018 lite TTTEEE + low-ℓ TT + low-ℓ EE + DESI DR2 BAO (13 points);
- PDL = PD + Planck 2018 lensing reconstruction.
- **Statistic:** $Q = \Sigma\chi^2 + ((A_{planck} - 1)/0.0025)^2$. The statistic, written down with a dated hash before the runs, is T_B = Q(D-B) − Q(free). Bands: $T_B \le 4$ compatible; $4 < T_B < 9$ tension; $T_B \ge 9$ negative in the declared model.
- **Free reference:** the refined profile point at $\Gamma = 0$ of section 4. It is a decaying-dark-matter run with the decay

switched off, equivalent to ΛCDM within 0.011 in Q.

data	Q found, D-B	Q found, free	T_B	free ω_c/ω_b
PD	1018.007	1017.837	**0.17**	5.221
PDL	1027.828	1027.600	**0.23**	5.207

- **The values are numerical estimates from the minima found, not certified global minima.**
- The two starts of the PD constrained fit differ by 0.30 in Q, more than the reported penalty.
- The lowest free ΛCDM minima found in other runs (`corse_6/`: L0_PDL; DD_PD, which ended at Γ = 0) give T_B = 0.26 and 0.24.
- **Where the cost comes from:** the high-ℓ Planck likelihood worsens by 0.9-1.5; DESI improves by 0.5-0.7.
- **Cross-check:** an earlier CAMB fit (Planck + DESI DR2) gave +0.04.
- **Limits.**
- Σm_ν was fixed.
- The uncertainty of the free ratio is not quantified here.
- Nothing here derives 3√3.
- **Registration for future data:** `registrazioni/REGISTRATION_3SQRT3_EN.md`.

3. The RAR scale versus redshift (`codice/07_rar_scale_evolution.py`, `registrazioni/a0_redshift/`)

Data.

- The four MUSE-DARK III bins tabulated by Alcock (2026), digitised from figure 3 of Ciocan et al. (2026).
- The SPARC anchor 1.20 ± 0.02 ± 0.24 (amplitudes in units of 10^{-10} m s^{-2}).
- The confidence level of the grey error bars is not stated explicitly. The source quotes 95 % intervals for its full-sample fit and applies the same procedure to the bins. Both readings are therefore kept, c68 (1σ) and c95 (95 %), without choosing either on the basis of its fit quality.

Treatment. Flat background with Ω_m = 0.315, Ω_Λ = 0.685. The χ^2 treat the digitised bins as independent; shared modelling and calibration uncertainties are not fully represented by this approximation.

model and fitted parameters	c68: parameters	c68: χ^2 (dof)	c95: χ^2
constant: A	A = 2.279	28.83 (3)	110.8
Hubble rate, A·H(z)/H_0	A = 1.390 ± 0.032	5.48 (3), p = 0.14	21.04
inverse age, A·t_0/t(z)	A = 1.069	15.33 (3)	58.9
power, A(1+z)^γ	γ = 0.78 ± 0.15 (0.83 ± 0.13 with SPARC)	1.16 (2)	4.5

What it shows. The four digitised bins favour an increasing scale. Among the three one-parameter laws, H(z) gives the lowest χ^2 (ΔAICc = 9.85 over the inverse-age law; non-nested models, so no conversion into σ). With the c95 reading H(z) is a poor fit, although better than the other two: relative preference and absolute adequacy are kept separate.

What it does not show.

- **Not unique.** Forms chosen after seeing the data, such as (1+z)^¾ (χ^2 1.20) or 1 + ln(1+z) (2.39), do as well or better.
- **H is not a power law.** Its effective logarithmic slope is 0.84 between z = 0 and 1 and 1.11 between the extreme bins.
- **Rescaling.** A redshift-independent multiplicative rescaling of all inferred a_0 values changes the amplitude but not the evolution. A common stellar-mass systematic need not propagate in exactly this way.
- **H and age are close in shape.** Once their amplitudes are fitted, the two laws differ by only 0.062 dex between z = 0.5 and 2.5.

Tension.

- At z ≈ 2 the H law with A = 1.39 gives $a_0 = 4.21$. On the galaxies of Genzel et al. 2017 discussed by Milgrom (2017), this would imply dark fractions well above the measured ones (an illustrative check).
- In KMOS3D the baryonic Tully-Fisher relation has a higher baryonic mass at fixed circular velocity at z ≈ 2.3 than at z ≈ 0.9. This is a qualitative tension with an increasing a_0 under $V_f^4 = G M_b a_0$, not a direct measurement of the evolution of a_0.

MM 2.42. The inverse-age law registered in MM 2.42 fits these bins less well than H(z), under the stated error assumptions. The original registration remains unchanged. Its reassessment is documented separately in version 2.42.1 of that record. H(z) is retained here as a phenomenological hypothesis for future testing: `registrazioni/REGISTRATION_A0_HUBBLE_EN.md`. The working registration is the Italian `registrazioni/a0_redshift/PREVISIONE_A0_HUBBLE_v2.md`.

4. Limits on uniform decay (`codice/08_cosmology_results.py`, `registrazioni/decadimento/`)

Model.

- Almost all cold dark matter decays at a constant rate Γ into dark radiation (CLASS dcdm, $\omega_{cdm} = 10^{-5}$ kept for the synchronous gauge).
- The comoving fraction lost between t_i and today is $F = 1 - \exp[-\Gamma(t_0 - t_i)]$. The table uses $t_i \approx 0$ and t_0 from CLASS; taking t_i at recombination changes none of the rounded limits.
- «Lifetime» is Γ^{-1}.
- **The decay products gravitate.** This analysis therefore does not constrain disappearance into a component postulated not to contribute to the shared gravitational field.

Profile (Γ fixed, the other parameters minimised; two starts near the best point; BOBYQA, rhoend 0.001):

Γ (km s^{-1} Mpc^{-1})	loss	ΔQ, PDL	ΔQ, PD
0	0	0	0
0.5	0.7 %	0.36	—
1.0	1.4 %	1.49	1.00
1.5	2.1 %	2.21	—
2.01	2.8 %	2.92	2.12
3.0	4.1 %	4.56	2.88

Result.

- **No preference for constant-rate decay** is found in the sampled profiles; their lowest value is at Γ = 0.
- **Bounds.** With the conventional likelihood-rise threshold ΔQ = 2.71 (monotone cubic interpolation), the nominal bounds are:
- PDL: Γ < 1.86, loss < 2.6 %, lifetime > 525 Gyr;
- PD: Γ < 2.74, loss < 3.8 %, lifetime > 357 Gyr.
- **These bounds are provisional.** Their frequentist coverage has not been calibrated, and the numerical convergence of the profiles is incomplete. In 6 of 10 points the two starts differ by more than the planned tolerance of 0.1 in Q (up to 0.54). A shift of 0.3 in ΔQ moves the bound by about 0.3 points of loss (PDL) and 0.6 (PD). This is a sensitivity, not an error bar.

Validations.

- ΛCDM and the decaying model at Γ = 0 agree within 0.011 in Q.
- The linear-versus-halofit comparison changes ΔQ(2.01) − ΔQ(0) by −0.04 at the tested points. This is a sensitivity check, not a validation of the nonlinear prescription for the decaying model.
- CLASS ΛCDM and CAMB ΛCDM differ by about 0.35 in Q; the shifts in ω_b, ω_c and H_0 are below 0.27 times the reference uncertainties.

Context. The nominal lifetime bounds are numerically larger than the published 246 Gyr (Alvi et al. 2022) and 288 Gyr

(Bucko, Giri and Schneider 2023). Different data sets, likelihoods and statistical constructions prevent a direct comparison of constraining power.

5. Clusters (`codice/09_rar_reading.py`)

- **The constraint is on mean mass densities.** The imposed relation concerns cosmological mean mass densities, ρ_c/ρ_b = 3√3. It does not by itself fix a particle-number ratio, a local density ratio or a separate mass budget for each halo.
- **The baryon fraction.** Excluding massive neutrinos, the baryon fraction of baryons plus cold dark matter is 1/(1 + 3√3) = 0.161.
- **X-COP.** In the five-system subsample with stellar masses (Eckert et al. 2022), the hydrostatic baryon fractions within R_{500} range from 0.141 to 0.165 for four systems. A2319 gives 0.185 and is flagged as strongly affected by hydrostatic bias. This is a rough mass-budget comparison; no likelihood analysis separates the fixed ratio from the ΛCDM value 0.157.
- **X-COP also finds that the galaxy RAR does not describe the acceleration relation in clusters.** Agreement of the integrated baryon budget does not resolve this separate problem.
- **Galaxy halos alone are not enough.** With f_gas = 0.12-0.14 and f_* = 0.015-0.025, the dark mass needed per unit stellar mass is 33-58. The spherical RAR estimate within 100 kpc gives 4.9-16 for stellar masses 3×10^{11}-3×10^{10} M⊙. So summing inner galaxy contributions is insufficient; this does not determine the full cluster mass distribution.

6. The hypothesis and the MM reading (interpretation)

Stated hypothesis.

- We hypothesise that the effective gravitating cold-dark-matter density represents a contribution from hidden-sector matter with its own internal interactions, distinct from the visible baryons counted in ω_b. Its particle masses, dimensionless couplings and thermal history remain unspecified. We separately impose ρ_c/ρ_b = 3√3.
- In MM terms the hidden sector is a «mirror», normal in its own laws, whose constants are read relative to ours. Which constants differ, and how, is left generic in this record.
- Under standard universal gravitational coupling, a thermally identical Standard Model copy is incompatible with radiation-density constraints: after electron-positron annihilation its photons and neutrinos would contribute ΔN_eff ≈ 7.4. Identical constants do not impose identical temperatures, and this argument does not directly apply to the selective radiation coupling postulated below. Collisions, cooling and possible disc formation require separate modelling.
- The worked example of section 9 satisfies two approximate screening inequalities for scattering and residual ionisation. It does not establish a cosmologically and astrophysically viable hidden sector. The fixed-ratio ΛCDM checks test that parametrisation, not all the properties of the generic hidden sector.

What is shared and what is not.

- The hidden light and internal forces do not act on us.
- The reading additionally postulates that hidden radiation does not source our cosmological expansion. Electromagnetic separation alone does not imply this property, and no covariant gravitational model implementing this selective coupling is supplied here.

How much and where.

- The amount is constrained by the mean ratio 3√3. Its spatial distribution is not specified by the bond.
- For the cosmological fits, the gravitating component is represented by standard stable CDM. This defines a restricted ΛCDM fit; it does not establish that the proposed hidden sector reproduces the required background and perturbation dynamics.

The RAR as two readings of one local object.

- For a spherical baryonic field, define L = 2GM_b(<r)/(c²r), S = r/R_H and a_MM = c²/(2R_H). Then L/S = g_bar/a_MM.
- The empirical RAR argument is y = g_bar/a_0 = (a_MM/a_0)(L/S). At z = 0, with the adopted H_0 and the SPARC normalisation, a_MM/a_0 = 2.73. The extra fraction of the total radial field is exp(−√y).
- MM interprets the baryonic and the additional contributions as two readings of the same object. This interpretation does not supply a dynamical mechanism for the RAR, and it does not remove the gravitational interaction assumed in the cosmological calculation. A quantitative account of the RAR's shape and scatter

remains open.

Density form.

- Outside a concentrated spherical baryonic mass, the leading deep-RAR equivalent dark density is $\rho_D \simeq \sqrt{[a_0 \bar{\rho}_b(<r)/(12\pi G r)]} = \kappa\sqrt{[\bar{\rho}_b(<r)\, \rho_{crit}/S]}$.
- Here $\bar{\rho}_b(<r) = 3M_b/(4\pi r^3)$ is the mean enclosed baryonic density, not the local density at r.
- $\kappa = 1/3$ exactly for $a_0 = a_{MM}$, and 0.20 for the measured a_0.
- This is an asymptotic rewriting of the assumed RAR in the stated geometry, not an independent density law.

A finite reservoir (conditional).

- Suppose an isolated system is additionally assigned a finite dark reservoir $M_{D,max}$, and its outer field is treated as spherical with a concentrated baryonic mass M_b. Then the adopted RAR reaches that reservoir at $r_{es} = \sqrt{(GM_b/a_0)}/\ln(1 + 1/q_{max})$, with $q_{max} = M_{D,max}/M_b$.
- Assigning $M_{D,max} = 3\sqrt{3}\cdot M_{b,collected}$ requires a separate prescription for the collected baryons.
- **Example:** $M_b = 6\times10^{10}$ M⊙ and 1.6×10^{11} M⊙ collected give $r_{es} \approx 120$ kpc, and 113 km s^{-1} at 300 kpc instead of
- These numbers assume a saturated reservoir and only 6×10^{10} M⊙ of gravitating baryons within 300 kpc; additional circumgalactic baryons would change them.
- A comparison with the weak lensing of isolated galaxies (Mistele et al. 2024) is possible only once the reservoir is defined independently.

Normalisation. The literal value $a_{MM} = cH_0/2 = 3.27\times10^{-10}$ m s^{-2} is 2.73 times the SPARC value. In this record a_0 is taken from data. A toy algebraic rescaling of the local vertical field near the Sun gives about 66 and 92 M⊙ pc^{-2} for the empirical and the literal normalisations. This is not a self-consistent Galactic mass model and is not used as a quantitative test.

7. Problems this reading tries to connect

The motivation is to investigate whether one hidden-sector interpretation can connect several dark-matter puzzles. The connections currently have different statuses: imposed numerical hypotheses, phenomenological fits and qualitative assumptions. Other research programmes also address combinations of these puzzles. Today this is an aim of unification, not four consequences derived from one model.

The case for this reading rests on consilience: one interpretation connecting puzzles that are usually treated separately. That case is only as strong as the links that become quantitative; today two of the four do.

puzzle	quantitative status in this record	related approaches
Dark-to-baryon density ratio	Fixed numerical hypothesis, not derived: 5.196; retrospective consistency check (T_B 0.17-0.23) and registered future test	Asymmetric and mirror dark matter: model-dependent quantitative abundance relations (Kaplan, Luty and Zurek 2009; Berezhiani 2008)
Cosmological scale of a_0	Phenomenological H(z) fit: lowest χ^2 among three specified one-parameter laws, but inadequate under the c95 reading; the literal normalisation is high by 2.73	Milgrom's cosmological acceleration connection (Milgrom 1999); Verlinde's de Sitter construction (equivalent scale $cH_0/6$), with different assumptions about evolution
Absence of conventional direct-detection signals	Absence of non-gravitational interactions with visible matter is assumed; no detection-rate calculation is supplied	Hidden and mirror sectors with absent or weak portal interactions
CDM-like cosmology and galaxy acceleration relations	CDM behaviour is assumed and the empirical RAR is imposed; their dynamical connection is not derived	AeST (Skordis and Złośnik 2021); superfluid dark matter (Berezhiani and Khoury 2015); ΛCDM galaxy-formation models (Ludlow et al. 2017); some dissipative hidden-sector models (Chashchina, Foot and Silagadze 2017)

Two quantitative hypotheses are examined here: the fixed density ratio and the H(z) evolution law. Neither is derived from the hidden-sector construction. The decay analysis of section 4 is an additional constraint on one phenomenological model, not an explanation of dark-matter stability.

8. Part 1 — what the MM algebra provides for a hidden sector (`codice/01_algebra_sectors.py`, `06_light_separation.py`)

Kept from an earlier unpublished draft. The derivations are in `documenti/parte1_algebra/astra/`; the independent checks are in `documenti/parte1_algebra/verifiche/` (both in Italian). The modules recompute the stated quantities on three banks.

statement	status
A declared translation protects closed modes	commutator residuals at numerical precision (largest element below 4×10^{-15}); exact invariance is an algebraic claim under the stated definitions (`ESITO_12`)
Closed modes weigh like all the others	a partition of unity ($\Sigma\varepsilon_i = H$)
Invisibility to our light must be assumed	every closed cycle responds to an edge connection at second order (minimum response 0.0036-0.033)
A whole neutral copy is compatible with weight; protecting only the cycles of one copy fails	`ESITO_13`
The eight-state scaffold of MM 2.29 has two equivalent copies of equal mass	derived in the scaffold (`ESITO_18`)
A conserved common number not carried by photons	forbids decay into photons alone for a pair that carries a non-zero total of that number
Separation of the two lights	obtained on disconnected light networks in the declared class of local operators; checked on 60 random graphs; grouping channels is not enough (the reading returns, up to 25.5 %); a small link between the photons reopens communication at large distance

- **Equal weights in the algebra** do not determine cosmological abundances or derive $3\sqrt{3}$. Identifying them with physical gravitational masses is an interpretation.
- **Left out:** the quantitative 1 TeV candidate of the earlier draft. It is superseded by the generic reading of section 6.

9. A worked special case: one common scale factor (`codice/10_hidden_sector_scaling.py`)

The map. Consider $\ell_{eff} = \ell'/A$, $t_{eff} = t'/A$ and $m_{eff} = A\,m'$, with c, $\hbar$ and α unchanged. Define $x = T'_\gamma/T_\gamma$, so that the effective photon temperature ratio is Ax. Then $\rho_{eff} = A^4\rho'$ and $(\sigma/m)_{eff} = A^{-3}(\sigma/m)'$. Applying the map physically also requires a prescription for the shared geometry and the gravitational coupling.

A screening region. An illustrative screening imposes:

- $(\sigma/m)_{eff} \le 1\ \mathrm{cm^2\,g^{-1}}$, using a cold neutral-hydrogen estimate;
- a residual ionised fraction $x_e \le 0.01$, using an approximate recombination scaling.

These choices give $A \gtrsim 850$ and $x \lesssim 520/A^3$. Hidden neutrinos and radiation are then negligible. This defines a formal screening region, not an established viable parameter window.

- The agreement with the Cyr-Racine-Sigurdson estimate concerns the scaling; their formula is itself uncertain by up to a factor of ten.
- Three-body reactions (important throughout the region; Gurian et al. 2022), the hidden nucleosynthesis, ionisation during collisions and collective plasma effects (Lasenby 2020; DeRocco and Giffin 2025) are not computed.

Content.

- At $A = 1000$ and $x \approx 5.2\times10^{-7}$, the illustrative upper edge, $\eta' \approx 0.023$, about 44 photons per baryon. This varies across the region.
- A copy with our baryon-to-photon ratio fails the adopted screening in the example considered. No observationally viable range is established.
- The hidden proton reads as $A\cdot m_p$ (0.94 TeV at $A = 1000$), an illustrative conversion.

This is why the record keeps the relative constants generic.

10. How this work can turn out wrong

Two quantitative hypotheses have predeclared failure thresholds for prospective tests. The dated analysis specifications must be completed before the relevant new results are inspected.

- **The ratio.** If DESI final BAO + Simons Observatory + complementary Planck, analysed as registered with Σm_ν free, give $T_B \geq 9$, the fixed ratio $3\sqrt{3}$ fails in the declared model.
- The sensitivity of this combination has not been forecast here.
- In a Gaussian approximation, ratios of 5.30 or 5.10 would nominally reach $T_B = 9$ if $\sigma(\omega_c/\omega_b)$ were about 0.035 or 0.032. Actual detection power requires a dedicated forecast.
- **The shape of $a_0(z)$.** If the registered global shape test gives $p < 0.0027$, with one free normalisation per homogeneous sample shared across its redshift bins and the declared treatment of systematics, the specified Hubble-rate law fails that test.
- These outcomes concern the specified quantitative hypotheses within their declared assumptions.
- **Additional findings would challenge specified versions or assumptions.** Evidence for decay would challenge the stable-CDM version; merger dynamics could challenge a specified collisionless-halo realisation; microphysical calculations could rule out specified hidden-sector realisations. These are not additional registered failure thresholds.

11. What is not claimed

- **$3\sqrt{3}$ is not derived here.** It is a hypothesis of MM, consistent with present CMB + BAO data and registered for future data.
- **The RAR is not derived.** Its form and a_0 are taken from data; the literal MM normalisation is off by 2.73.
- **The hidden sector's dynamics is not computed.** Its motion as cold dark matter is assumed. The non-gravitating hidden radiation is a postulate without a covariant model.
- **The Bullet cluster is not computed.** A predominantly collisionless component is the intended qualitative interpretation, not a demonstrated merger solution.
- **The decay bounds are nominal and provisional,** for one model only.
- **No observational confirmation of MM is claimed.**
- **Numerical proximity is not treated as a derivation or as observational confirmation of MM.**

12. How the record was made

- **The author.** The framework and its premises are the author's. His words that started these calculations are in `WHERE_THIS_COMES_FROM.md`, with a translation. No result depends on them.
- **Sections 2-6 and 9.** The calculations and code are by Claude (Anthropic), an AI assistant. The working protocols are in `registrazioni/` (Italian, kept as written); they come with SHA-256 content hashes and dated records. A hash identifies a file's contents; any claim of prior registration relies on the documented chronology.
- **Cross-checks.** The archive includes cross-checks by a second AI assistant, Astra (GPT-6, in a Codex session), with their scope, corrections and unresolved points documented in `documenti/note_di_verifica/` (Italian). Claude re-verified the new numbers with code. These checks are not independent observational replication or human peer review.
- **Section 8 (Part 1).** The derivations are by Astra, checked by Claude with independent code (modules 01 and 06 are Claude's code).
- **Withdrawn statements are recorded,** not hidden: for example an early «3σ» for a non-nested comparison, a 3 % bound on condensed hidden matter, and a coincidence with $1 + \Omega_\Lambda$.
- **No human referee has reviewed this record.**

13. Reproduction

```
python riproduci.py          # about 1.5 minutes on one CPU
python riproduci.py --full   # also re-runs the slow optimisation of module 01
```

- **Requirements:** Python 3.11 or later with numpy, scipy and sympy. No network access.
- **Output:** `risultati/`.
- **The cosmological minimisations are not re-run by this command.**
- Their stored minima are in `dati/risultati_cosmologici/`:
- `corse_6/`: first runs, L0 = ΛCDM, DD = decay with Γ free, DF = Γ = 2.01;

- `profilo/`: profile points, R0, R2;
- `vincolo_3r3/`: D-B;
- `camb_confronto/`: CAMB ΛCDM.
- The scripts are in `codice/corse_cosmologiche/`.
- **To re-run them** you need Cobaya 3.6.2, CLASS 3.4.1 (classy) and the Planck 2018 likelihoods (`cobaya-install`). The scripts carry the paths of the original Linux machine (WSL). Adjust `packages_path` and the DESI data path in `desi_dr2_bao.py`; the DESI files are in `dati/desi_dr2/`, identical by SHA-256.
- **Run time:** one profile point takes 15-25 minutes on two threads.

14. Contents

path	what
`README.md`	this text
`LEGGIMI_IT.md`	summary in Italian
`WHERE_THIS_COMES_FROM.md`	the author's words, with translation (optional)
`riproduci.py`	reproduction of the included calculations and checks on archived results
`codice/01_*.py`, `06_*.py`	Part 1: algebra of hidden sectors; separation of light
`codice/07_*.py` … `10_*.py`	$a_0(z)$; cosmological results; RAR reading and clusters; one-factor special case
`codice/corse_cosmologiche/`	Cobaya/CLASS scripts used for the minimisations, and the DESI DR2 likelihood
`dati/`	MUSE-DARK III bins (Alcock 2026), DESI DR2 BAO mean and covariance, stored minima
`risultati/`	JSON and logs of the modules
`registrazioni/`	working protocols (Italian) and the two registrations in English
`documenti/`	verification notes (Italian) and the Part 1 derivations
`fonti/SOURCES.md`	references
`IMPRONTE.sha256`	SHA-256 of every file

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